

Homework 2. Markov decision problems

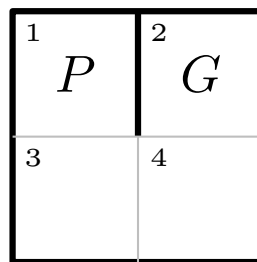


Figure 1: A taxi driver must pick up a passenger standing in the cell marked with “P” and drop it in the cell marked with “G”.

Consider a taxi driver moving in the grid-world environment of Fig. 1. The driver must pick up a passenger (the passenger is in the cell marked with “P”) and drop her at the cell marked with “G”.

At each step, the agent may move in any of the four directions—up, down, left and right. Movement across a *grey* cell division succeeds with a 0.8 probability and fails with a 0.2 probability. Movements across black cell divisions are not possible. When the movement fails, the driver remains in the same cell.

To pick up the passenger, the taxi driver simply needs to stand in the corresponding cell. In other words, as soon as the taxi driver stands on the cell marked with “P”, you may consider that the passenger is in the car. For simplicity, assume that once the passenger is in the taxi, it remains there forever—the taxi driver should, therefore, go to “G” and not leave.

Exercise 1.

- Identify the state space, $\mathcal{X}$, and the action space, $\mathcal{A}$, for the MDP.
- Write down the transition probability matrix for the action “down” and a (possible) cost function for the MDP. Make sure that the cost function is as simple as possible and verifies

$c(x, a) \in [0, 1]$ for all states $x \in \mathcal{X}$ and actions $a \in \mathcal{A}$. Note, in particular, that the cost should depend only on the taxi driver completing the delivery of the passenger.

- (c) Compute the cost-to-go function associated with the policy in which the taxi driver always goes down, using a discount $\gamma = 0.9$. You can use any software of your liking for the harder computations, but should indicate all other computations.